Foundations of Lasso regression

[Lasso regression]

$$L(\beta) = \|y - X\beta\|^2 + \lambda\|\beta\|_1$$

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where q i {\displaystyle q_{i}} is specified below and the "prime" symbol stands for transpose. The first fraction represents relative accuracy, the second fraction relative simplicity, and λ {\displaystyle \lambda } balances between the two.

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where y ~ 0 = y − X β 0 {\displaystyle {\tilde {y}}_{0}=y-X\beta _{0}} . If b = b OLS {\displaystyle b=b_{\text{OLS}}} when R 2 {\displaystyle R^{2}} is computed, then the diagonal elements of R ⊗ {\displaystyle R^{\otimes }} sum to R 2 {\displaystyle R^{2}} . The diagonal R ⊗ {\displaystyle R^{\otimes }} values may be smaller than 0 or, less often, larger than 1. If regressors are uncorrelated, then the i t h {\displaystyle i^{th}} diagonal element of R ⊗ {\displaystyle R^{\otimes }} simply corresponds to the r 2 {\displaystyle r^{2}} value between x i {\displaystyle x_{i}} and y {\displaystyle y} . A rescaled version of the adaptive lasso of can be obtained by setting q adaptive lasso , i = | b OLS , i − β 0 , i | {\displaystyle q_{{\mbox{adaptive lasso}},i}=|b_{{\text{OLS}},i}-\beta _{0,i}|} . If regressors are uncorrelated, the moment that the i t h {\displaystyle i^{th}} parameter is activated is given by the i t h {\displaystyle i^{th}} diagonal element of R ⊗ {\displaystyle R^{\otimes }} . Assuming for convenience that β 0 {\displaystyle \beta _{0}} is a vector of zeros,

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min β ∈ R p { 1 N ■ y ~ − X β ■ 2 2 + λ 1 + η ■ β ■ 1 } , {\displaystyle \min _{\beta \in \mathbb {R} ^{p}}\left\{{\frac {1}{N}}\left\|{\tilde {y}}-X\beta \right\|_{2}^{2}+{\frac {\lambda }{1+\eta }}\|\beta \|_{1}\right\},}

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min β ∈ R p { 1 N ■ y − X β ■ 2 2 + λ ∑ j = 1 p ϑ (β j 2) } , {\displaystyle \min _{\beta \in \mathbb {R} ^{p}}\left\{{\frac {1}{N}}\left\|y-X\beta \right\|_{2}^{2}+\lambda \sum _{j=1}^{p}\vartheta (\beta _{j}^{2})\right\},}

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That is, if regressors are uncorrelated, λ {\displaystyle \lambda } again specifies the minimal influence of β 0 {\displaystyle \beta _{0}} . Even when regressors are correlated, the first time that a regression parameter is activated occurs when λ {\displaystyle \lambda } is equal to the highest diagonal element of R ⊗ {\displaystyle R^{\otimes }} . These results can be compared to a rescaled version of the lasso by defining q lasso , i = 1 p ∑ l | b OLS , l − β 0 , l | {\displaystyle q_{{\mbox{lasso}},i}={\frac {1}{p}}\sum _{l}|b_{{\text{OLS}},l}-\beta _{0,l}|} , which is the average absolute deviation of b OLS {\displaystyle b_{\text{OLS}}} from β 0 {\displaystyle \beta _{0}} . Assuming that regressors are uncorrelated, then the moment of activation of the i t h {\displaystyle i^{th}} regressor is given by

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In statistics and machine learning, lasso (least absolute shrinkage and selection operator; also Lasso, LASSO or L1 regularization) is a regression analysis method that performs both variable selection and regularization in order to enhance the prediction accuracy and interpretability of the resulting statistical model. The lasso method assumes that the coefficients of the linear model are sparse, meaning that few of them are non-zero. It was originally introduced in geophysics, and later by Robert Tibshirani, who coined the term. Lasso was originally formulated for linear regression models. This simple case reveals a substantial amount about the estimator. These include its relationship to ridge regression and best subset selection and the connections between lasso coefficient estimates and so-called soft thresholding. It also reveals that (like standard linear regression) the coefficient estimates do not need to be unique if covariates are collinear. Though originally defined for linear regression, lasso regularization is easily extended to other statistical models including generalized linear models, generalized estimating equations, proportional hazards models, and M-estimators. Lasso's ability to perform subset selection relies on the form of the constraint and has a variety of interpretations including in terms of geometry, Bayesian statistics and convex analysis. The LASSO is closely related to basis pursuit denoising.

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where ϑ (γ) {\displaystyle \vartheta (\gamma)} is an arbitrary concave monotonically increasing function (for example, ϑ (γ) = γ {\displaystyle \vartheta (\gamma)={\sqrt {\gamma }}} gives the lasso penalty and ϑ (γ) = γ 1 / 4 {\displaystyle \vartheta (\gamma)=\gamma ^{1/4}} gives the ■ 1 / 2 {\displaystyle \ell ^{1/2}} penalty). The efficient algorithm for minimization is based on piece-wise quadratic approximation of subquadratic growth (PQSQ).

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min β 0 , β { ■ y – β 0 – X β ■ 2 2 } subject to (1 – α) ■ β ■ 1 + α ■ β ■ 2 2 ≤ t , where α = λ 2 λ 1 + λ 2 . {\displaystyle {\begin{aligned}\min _{\beta _{0},\beta }\left\{\left\|y-\beta _{0}-X\beta \right\|_{2}^{2}\right\}&{\text{ subject to }}(1-\alpha)\|\beta \|_{1}+\alpha \|\beta \|_{2}^{2}\leq t,\\&{\text{ where }}\alpha ={\frac {\lambda _{2}}{\lambda _{1}+\lambda _{2}}}.\end{aligned}}}

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Lasso can set coefficients to zero, while the superficially similar ridge regression cannot. This is due to the difference in the shape of their constraint boundaries. Both lasso and ridge regression can be interpreted as minimizing the same objective function

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For p = 1 {\displaystyle p=1} , the moment of activation is again given by λ ~ lasso , i = R 2 {\displaystyle {\tilde {\lambda }}_{{\text{lasso}},i}=R^{2}} . If β 0 {\displaystyle \beta _{0}} is a vector of zeros and a subset of p B {\displaystyle p_{B}} relevant parameters are equally responsible for a perfect fit of R 2 = 1 {\displaystyle R^{2}=1} , then this subset is activated at a λ {\displaystyle \lambda } value of 1 p {\displaystyle {\frac {1}{p}}} . The moment of activation of a relevant regressor then equals 1 p 1 p B p B 1 p B = 1 p {\displaystyle {\frac {1}{p}}{\frac {1}{\sqrt {p_{B}}}}p_{B}{\frac {1}{\sqrt {p_{B}}}}={\frac {1}{p}}} . In other words, the inclusion of irrelevant regressors delays the moment that relevant regressors are activated by this rescaled lasso. The adaptive lasso and the lasso are special cases of a '1ASTc' estimator. The latter only groups parameters together if the absolute correlation among regressors is larger than a user-specified value.

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Using X ■ X = I {\displaystyle \ X^{\intercal }X=I\ } and the ridge regression formula: β ^ = (X ■ X + N λ I) – 1 X ■ y , {\displaystyle \ {\hat {\beta }}={\Bigl (}\ X^{\intercal }X\ +\ N\ \lambda \ I\ {\Bigr)}^{-1}X^{\intercal }y\ ,} yields: